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Comparative Evaluation of Different Kinetic and Diffusible Models to Identify the Most Accurate Predictor of Shale Swelling Behavior

Massab Hasnain and Yousuf Bin Arif, Department of Petroleum Engineering, NED University of Engineering & Technology, University Road, Pakistan

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Abstract

Shale formation contains clay minerals that readily absorb water, which leads to swelling when exposed to water-based drilling fluid (WBDF). This study focuses on analyzing the swelling characteristics of shale collected from Murree, located in northern Pakistan, and identifying the most appropriate kinetic and diffusible models to predict swelling behavior by comparing the experimental swelling with model-predicted swelling. Additionally, QXRD and SEM analyses were conducted to determine the mineralogical composition and microstructural features of the Murree shale formation. Experimental swelling percentages were measured using a linear dynamic swelling meter (LDSM). Swelling was continuously monitored in LDSM for 24 hours with data recorded at 30-second intervals. Five kinetic and diffusible models, namely, the first-order exponential association model, the pseudo-2nd-order kinetic model, Peleg's model, Higuchi model, and Peppas model, were applied to predict the swelling behavior. Statistical analysis, including absolute error, percentage relative error, mean absolute error (MAE), and absolute percentage relative error (APRE), was conducted to evaluate model performance. The results showed that the pseudo-2nd-order kinetic model performed the best with the lowest MAE of 0.291% and the highest accuracy 96%, while the Higuchi model performed the poorest, with the highest MAE of 1.116% and a lower accuracy of 84%. The first-order exponential association model, Peleg's model, and the Peppas model exhibited moderate performance. However, Peppas' model performed slightly better than the first-order exponential association model and Peleg's model. Furthermore, the analysis plotted a cross-plot between experimental swelling and predicted swelling. This model-based approach offers a reliable tool for anticipating shale swelling and supports improved decision-making for managing wellbore instability during drilling operations.

Keywords: Shale swelling, WBDF, Kinetic and diffusible models, Swelling prediction

Introduction

Shale is fine-grained sediments primarily composed of quartz, silt, and clay, with grain size generally smaller than 32 micrometers (i.e., <0.032mm). The characterization of shale and its tendency to break into thin, millimeter-scale layers parallel to sedimentary planes is commonly referred to as shaly fissility [1]. Clay

consists of clay minerals and has a particle size less than 2 micrometers (i.e., $<0.002\text{mm}$) [2]. Shale is a source rock and contains 40% of the stratigraphy beneath the earth's surface, and shale is responsible for over 70% of the wellbore stability issues [3, 4]. The origin of shale can be classified as authigenic and detrital. Authigenic minerals provide paleoclimate, while detrital minerals identify the provenance of the material [5]. Authigenic are those formed or generated in place within the sedimentary rock after the deposition, including rock constituents and minerals that crystallized locally at the spot [6]. Clay minerals play a very important role in shale and primarily consist of smectite, illite, and chlorite, and also a mixture of smectite and illite [7]. Shale instability in the wellbore is primarily attributed to the presence of clay minerals [8, 9]. Wellbore instability in shale can be categorized into mechanical, chemical, electrical, and thermal effects [10]. Wellbore instability is a major problem in shale formation due to both mechanical and chemical factors, leading to hole pack-off, bridging, stuck pipe, and non-productive time [11, 12]. Inadequate hole cleaning, poor mud properties, and swelling due to fluid interaction are the major causes of instability [13–15]. Mechanical effects include drill string interaction with shale formations, erosive fluid forces, pressure fluctuations, and the disturbance of in-situ vertical and horizontal stresses, and overpressure presence contributes to shale instability during drilling operation [16]. However, shale creates problems such as shrinking, swelling, strength reduction, and hydration due to water-attracting properties of clay particles, and these are due to chemical effects caused by intermolecular attraction between ionic pore fluids and clay particles within the shale matrix, which are further influenced by the composition of drilling fluid [17]. The swelling mechanism is classified into two main mechanisms: crystalline and osmotic swelling. During crystalline swelling, water molecules adhere to the crystal surface through hydrogen bonding, forming a quasi-crystalline structure that increases the c-spacing. Osmotic swelling occurs when a high concentration of cations between clay layers draws water osmotically, resulting in greater expansion; however, significant osmotic swelling is primarily exhibited by sodium-montmorillonite [18]. Smectite, a highly reactive clay mineral, swells by adsorbing water within its layers and features a structural composition of one octahedral layer enclosed by two tetrahedral layers (T:O:T). It has a high swelling potential, and the tendency of sodium-rich smectite to swell often leads to wellbore instability during drilling operations [10]. Shale swelling destroys the pore structure and pore connectivity, leading to a reduction in the permeability, and the combined effect of water and shale is influenced by water salinity, confined pressure [19, 20]. High salinity water films exist in shale pores and low salinity fluid injection increases imbibition due to osmotic pressure; however, water movement in clay minerals is controlled by water salinity and clay swelling reduces permeability and clay fraction initial water content mainly influence the swelling potential of shale and confined pressure and water imbibition cause greater swelling resulting in blocking shale gas flow [21]. This study evaluates the performance of five established kinetic and diffusible models, such as the first-order exponential association model, pseudo-2nd-order kinetic model, Peleg's model, Higuchi model, and Peppas model. Their predictions were compared with experimental swelling data obtained from a linear dynamic swelling meter test conducted on the Murree shale formation. To determine which model most accurately predicts shale swelling behavior, statistical and graphical-based analysis approaches were employed. At the end of the 19th century, Lagergren proposed the first-order exponential association model [22–24]. However, the pseudo-2nd-order kinetic model proposed in the middle of 80's became the most popular in 1999 after several experimental results by Ho and McKay [25]. Pseudo-2nd-order kinetic model gives the best correlation of the experimental data [26].

Models applied in this study

First-order exponential association Model

By using this model, we describe the kinetic process that typically occurs under non-equilibrium conditions. [27]. Equation 1 shows the differential form of the first-order exponential model equation. When we integrate the differential form of the equation using the initial parameter, it yields a first-order exponential association

model equation, shown in Equation 2, and then, Macky and Ho proposed a linear relationship by taking the natural logarithm on each side of the equation. Equation 3 represents the linear equation, in which K_1 is the slope [28]. Where S_t , S_e and K_1 are the swelling at a given time, swelling at equilibrium condition, and the corresponding kinetic rate constant, respectively [28].

$$\frac{dS_t}{dt} = K_1(S_e - S_t) \quad (1)$$

$$S_t = S_e(1 - e^{-K_1 t}) \quad (2)$$

$$\ln(S_e - S_t) = \ln S_e - K_1 t \quad (3)$$

Pseudo-second-order kinetic Model

The pseudo-2nd-order kinetic model is commonly applied to the adsorption process. Equation 4 displays the differential form of the pseudo-second-order equation. When this differential equation is integrated under the initial pseudo-second-order condition, it is converted into a linear form. In this linear form ($1/q_e$) represents the slope, and $(1/K_2 q_e^2)$ serves as the intercept and can be obtained by plotting (t/q_t) vs t [29]. Equation 5 can be rewritten as shown in Equation 6. Where q_t , q_e , and K_2 denote the adsorption quantity over time (t), swelling at equilibrium condition, and pseudo-second-order rate constant, respectively [29].

$$\frac{dq_t}{dt} = K_2(q_e - q_t)^2 \quad (4)$$

$$\frac{t}{q_t} = \left(\frac{1}{K_2 q_e^2} \right) + \left(\frac{1}{q_e} \right) t \quad (5)$$

$$q_t = \frac{q_e^2 K_2 t}{1 + q_e K_2 t} \quad (6)$$

Peleg's model

An non-exponential model involving two parameters, which was developed by Peleg in 1988. This model was used to study the absorption of water molecules through grains in food items was studied using this model [28]. Equation 7 presents the mathematical expression of Peleg's model. Where K_1 and K_2 are characteristic rate constants that are inversely associated with both the initial water absorption rate and the moisture content at equilibrium condition, respectively, S_t and S_0 denote the swelling at a given time (t) and initial swelling, respectively.

$$S(t) = S_0 + \frac{t}{K_1 + K_2 t} \quad (7)$$

Higuchi model

The mathematical expression of the Higuchi model is applied to explain how drugs are released from various types of matrices, and it is particularly effective for systems that are already swollen [30, 31]. Equation 8 can be rewritten as shown in Equation 9 for determining the value of M_t at each period. Where, M_t represents the quantity of drug released over a specific time, M_∞ represents the complete drug released recorded at the end of the period, and k is the Higuchi characteristic rate constant [32].

$$\frac{M_t}{M_\infty} = K t^{1/2} \quad (8)$$

$$M_t = M_\infty K t^{1/2} \quad (9)$$

Peppas model

This model is a modified form of the Higuchi model that replaces the square root term with an adjustable parameter of " n " [33]. The other parameter remains the same as those used in the Higuchi model. This model

is often termed the power law model, but this approach offers a more detailed explanation of how drugs are released from polymeric-based systems [30, 31]. Equation 10 shows the mathematical expression of the Peppas model, while Equation 11 is written in another form to determine the quantity of drug released over a specific time period (M_t).

$$\frac{M_t}{M_\infty} = Kt^n \quad (10)$$

$$M_t = M_\infty Kt^n \quad (11)$$

Murree Shale Formation Location and Characteristics

The Murree shale formation was deposited during the Miocene geological period. Figure 1a) illustrated the graphical location of the sample used in this research study, which was collected from the Hazara Kashmir Syntax situated in the northern region of Pakistan. When retrieved, the shale cuttings showed a reddish brown color due to the presence of hematite content and had a high moisture content [34].

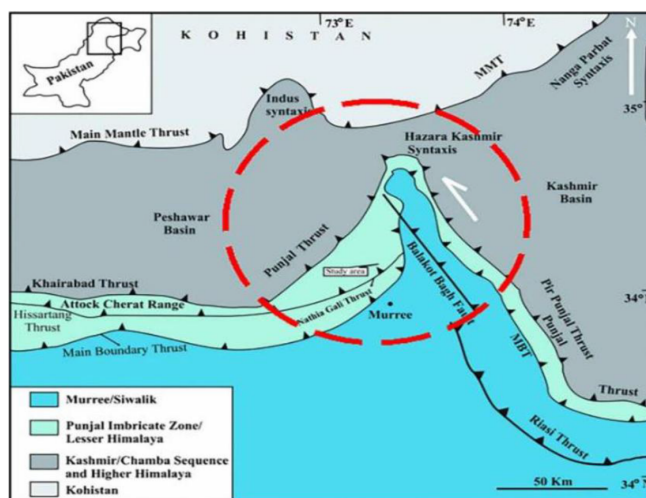


Figure 1a—Graphical location of the Murree shale formation [35]

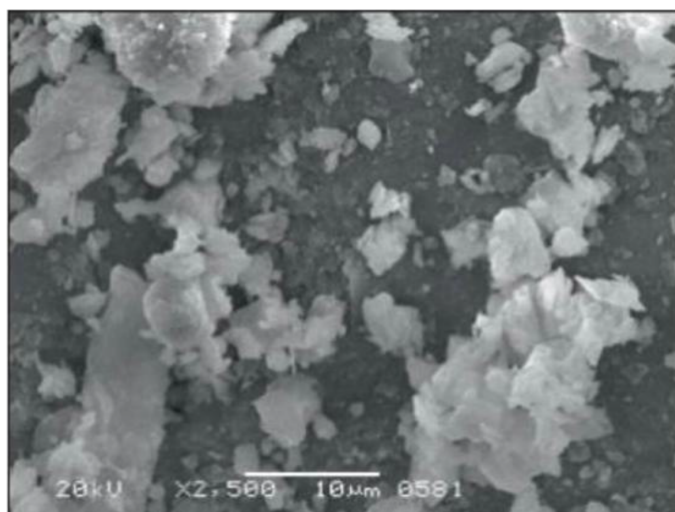


Figure 1b—SEM imaging of the Murree shale formation [35]

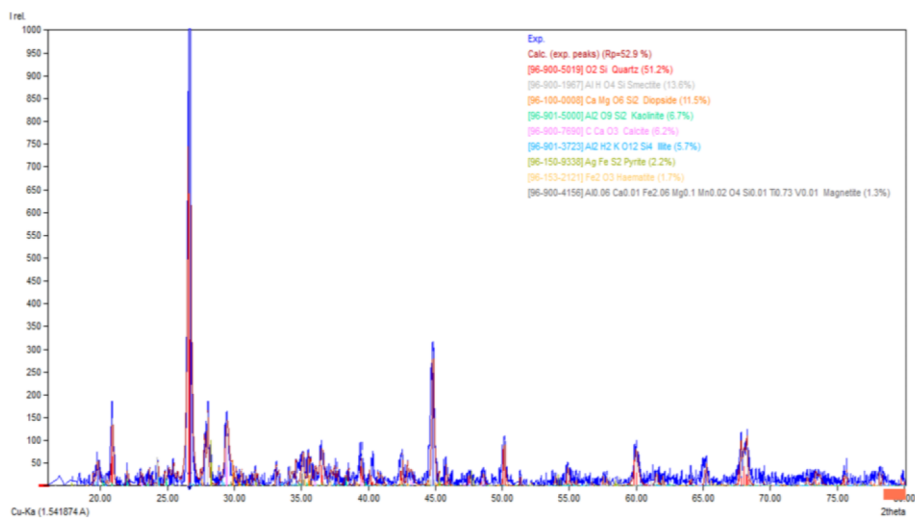


Figure 1c—XQRD evaluation of Murree shale formation [35]

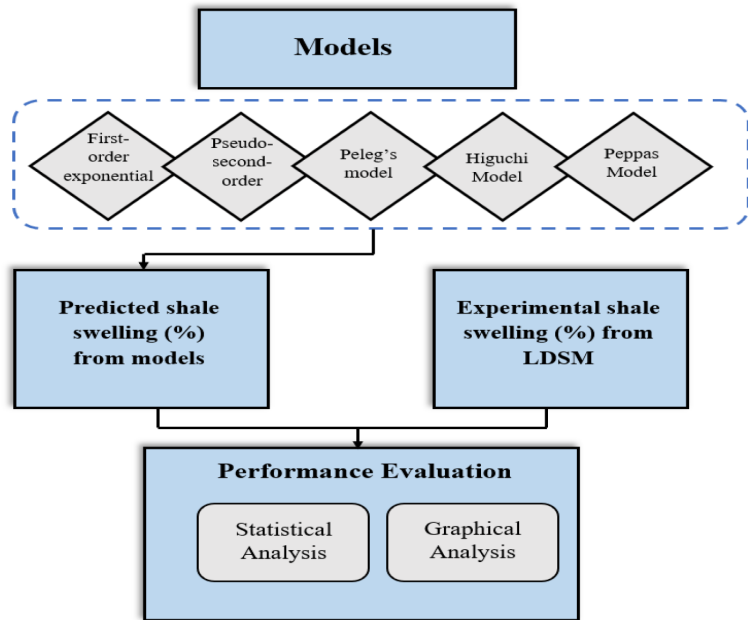
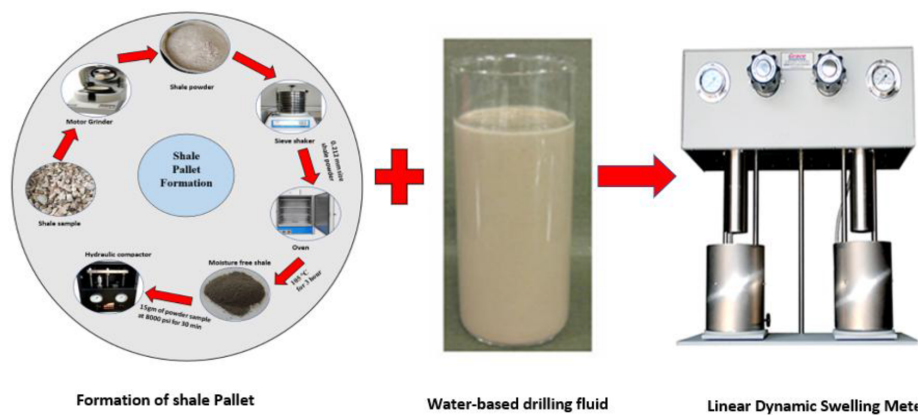


Figure 2—Sequential workflow for Experimental testing and Model fitting of shale swelling.

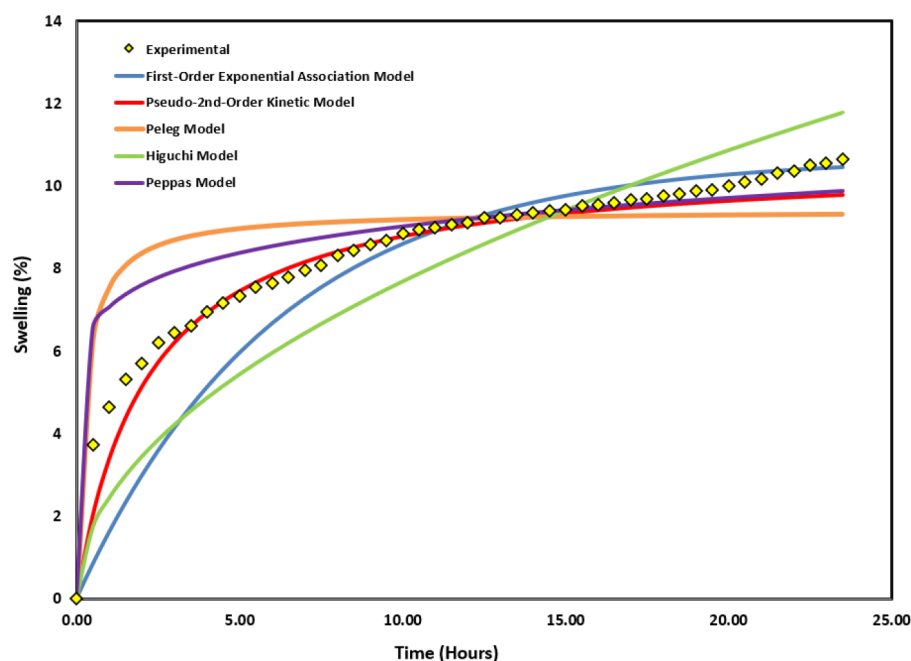


Figure 3—Experimental and model-predicted swelling behavior as a function of time.

The imaging of the Murree shale formation via through scanning electron microscope (SEM) is described in this research study, as shown in Figure 1b). Surface irregularities and folds observed in the examined shale formation reveal a crystalline structure. Additionally, the prominent sheet stacking present in the shale formation confirms the presence of smectite clay mineral. Furthermore, the extensive surface area of the shale sample can be attributed to the high abundance of microspores.

The Murree shale formation is mainly composed of a thick layer of mudstone and fine red shale. A diffractometer used a Cu filter and wavelength of 1.54060° was utilized to conduct the quantitative X-ray diffraction (QXRD) analysis of this formation. For this, a 0.5 g sample of shale was air dried in a moisture extraction oven [36]. The QXRD pattern of the Murree formation, as shown in Figure 1c). An interlayer spacing of kaolinite is illustrated by the first strong peak at $2\theta = 20.90^\circ$ [37]. According to Bragg's equation, the strongest peak was noted at $2\theta = 20.65^\circ$, corresponding to an interlayer spacing of 3.35 nm, which is characteristic of quartz [37]. Another notable peak appeared at $2\theta = 44.71^\circ$, associated with an interlayer separation of 2.02 nm, which falls within the range of montmorillonite [38]. Due to the higher volume of smectite, this clay mineral exhibits a greater swelling capacity [36].

The mineralogy of the Murree shale formation, as identified through QXRD analysis, is presented in Table 1. The minerals were categorized into four main groups: silicates, clay, iron ores, and carbonates. The composition of this formation was estimated to be approximately 63% silicates, 51.2% quartz, 5.1% iron minerals, and 26% clay, which is subsequently categorized into 13.5% smectite, 6.9% kaolinite, and 5.6% illite.

Table 1—Mineral composition of the Murree shale formation obtained from QXRD.

Mineral through QXRD	Mineral content (%)
Quartz	51.2
Diopside	11.5
Smectite	13.5
Kaolinite	6.9
Illite	5.6
Calcite	6.2
Pyrite	2.2
Hematite	1.7
Magnetite	1.3
Total	100

Methodology

Collecting Data for Modeling

To collect the data on swelling for modelling, it is necessary to find the swelling from an experiment. A shale sample is first prepared in pellet form for testing using a linear dynamic swelling meter (LDSM). The shale sample is collected from Murree, located in northern Pakistan, and turned into fine powder by using a motor grinder. The fine shale powder was then placed in a sieve shaker to obtain the required powder size (0.212mm). The powdered shale sample is then heated at 105 °C for 3 hours in an oven to remove moisture. After drying, 15 gm of moisture-free shale powder is placed in a hydraulic compactor, and a pressure of 8000 psi is applied for 30 min to form a shale pellet. Once the shale pellet is prepared, it is placed in the LDSM cell. Approximately 175 to 200 ml of WBDF is added to ensure contact with the base of the pellet. A transducer sensor is positioned on the top to record a vertical expansion of the sample. The displacement sensor records continuously the change in vertical height over 24 hours. The vertical height and corresponding time data are collected and used to calculate experimental swelling by using the following formula given in Equation 12.

$$\% \text{ Swelling} = \frac{h_t - h_o}{h_o} \times 100 \quad (12)$$

Model Performance Evaluation

After collecting the experimental swelling data concerning time from the Linear Dynamic Swelling Meter (LDSM) test, five kinetic and diffusible models were applied to determine which one best fits the experimental data. The models' equations, that is shown in equations 2, 6, 7, 8, and 10, were used to calculate the predicted swelling values. Absolute error was then determined by subtracting the predicted swelling from the experimental swelling. Subsequently, the mean absolute error (MAE) and absolute percentage relative error (APRE) were calculated using equations 13 and 14, respectively, for each model. The model with the lowest mean absolute error (MAE) was considered the best model because the lowest MAE indicated that the model's prediction is closer to the experimental values.

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^n |Y_{\text{exp}} - Y_m| \quad (13)$$

$$\text{APRE} = \frac{1}{N} \sum_{i=1}^n \frac{|Y_{\text{exp}} - Y_m|}{Y_{\text{exp}}} * 100 \quad (14)$$

Where, Y_{exp} represents the experimental swelling error values, Y_m denotes the predicted model swelling error values, and N is the no of observation points. For further analysis, plotting a graph of experimental

swelling vs predicted swelling, predicted swelling vs experimental time, absolute error vs experimental time, relative error vs experimental time, and a bar chart of MAE and APRE.

Result & Discussion

Tuning Parameters of Each Model

The characteristic rate constant for each model was determined through optimization by minimizing the error between experimental swelling and predicted swelling value. These characteristic rate constants were then utilized to plot the predicted swelling behavior curves for comparison with experimental swelling data, as shown in Table 2.

Table 2—Characteristic rate constant parameter for all different models.

Models	K_1	K_2	n	K
First-Order Exponential Association Model	0.163	-	-	-
Pseudo-2nd-Order Kinetics Model	-	0.043	-	-
Peleg's Model	0.026	0.106	-	-
Higuchi Model	-	-	-	0.227
Peppas Model	-	-	0.105	0.663

Comparison of Experimental and Modeled Swelling%

Figure 3 Displays the graphical representation of the Swelling (%) vs time (hours). It illustrates that the swelling predicted by the pseudo-2nd-order kinetic model approximately exhibits similar behavior observed in the experimental swelling data obtained from the linear dynamic swelling meter (LDSM) test. Additionally, it can be observed that the swelling predicted by the Higuchi model and the first-order exponential association model is higher compared to both the experimental and other swelling models, with the Higuchi model showing the highest swelling overall. Initially, Peppas' model predicts a higher swelling than experimental swelling; however, as time progresses, it aligns with the experimental swelling, but after some time, it starts below the experimental swelling.

Analysis of Models using Graphical Metrics

Figure 4 is a cross plot between experimental swelling and predicted swelling from different kinetic and diffusible models. The straight line in the graph represents the perfect model line, where predicted values exactly match the experimental values. The closer a model's data points are to the perfect line, the more accurate the prediction of swelling models is. It is demonstrating that the pseudo-2nd-order kinetic model shows the best fit, as its data points lie closest to the perfect model line, indicating the best predicted swelling model. In contrast, Higuchi and Peleg's model shows the greatest deviation from the perfect model line, especially at the highest data point, suggesting it is less reliable. As the swelling increases, the pseudo-2nd-order kinetic model aligns with the perfect model line. The first-order exponential association model and the Peppas model swelling exhibit a moderate deviation from the perfect model line. Near the end of the data points, all model lines appear to align more closely with the perfect model line, although the Higuchi model still shows the largest deviation, highlighting its poor performance.

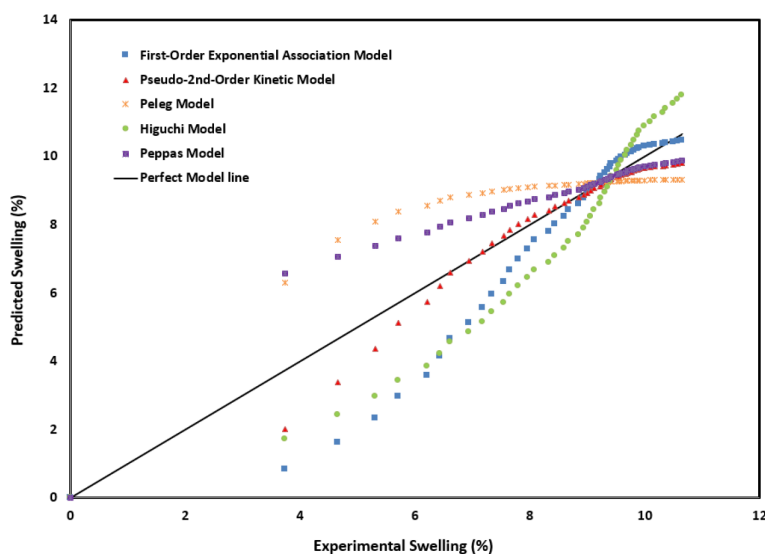


Figure 4—Cross-plots of experimental swelling and predicted swelling.

Statistical Error Evaluation

Figure 5 displays the relative error of all swelling models at each of the experimental time intervals. It illustrates that the relative error lines of the pseudo-2nd-order kinetic model remain closer to the x-axis, indicating minimal deviation between experimental swelling and predicted swelling. In contrast, the first-order exponential association model and Peleg's model deviate more from the x-axis, reflecting higher relative error and positioning them as the least accurate models. Higuchi and Peppas models also show relative error near zero, but still do not perform as well as the Pseudo-2nd-order kinetic model, which is the most accurate among all models. Furthermore, when comparing the pseudo-2nd-order kinetic model with all other models, it consistently demonstrates the best accuracy in terms of relative error, while the first-order exponential association performs the poorest in predicting shale swelling behavior.

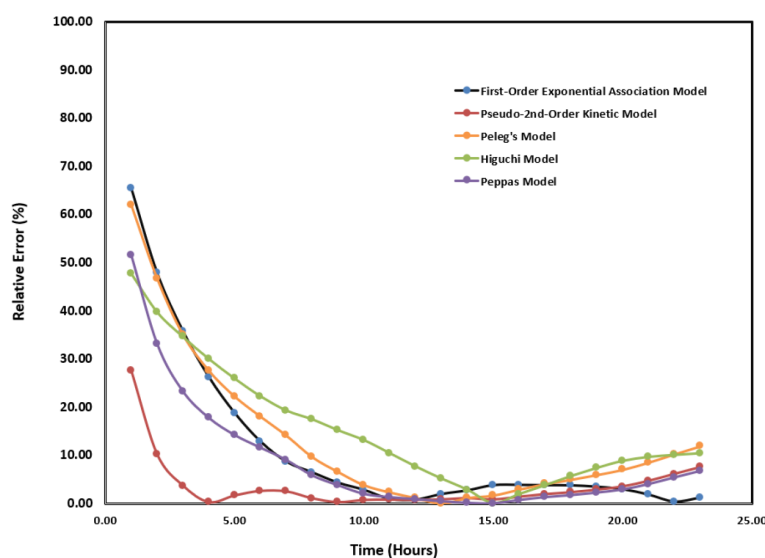


Figure 5—Relative Error variation with time for all models.

Figure 6 Shows the absolute error plot for different kinetic and diffusible models. The absolute error for each model exhibits nonlinear behavior. Among all models, the pseudo-2nd-order kinetics model shows the lowest maximum absolute error of 1.282%, indicating a minimal deviation from the experimental swelling obtained from the linear dynamic swelling meter. In contrast, the first-order exponential model and Peleg's

models display the highest deviation, with maximum absolute error of 3.046% and 2.887% respectively, which is approximately twice as large as that of the pseudo-second-order kinetics model. The Higuchi and Peppas model also shows higher maximum absolute error than the pseudo-second order kinetic model, but remains lower than the first-order exponential model and Peleg's models. Initially, the first-order exponential model exhibits a higher absolute error, but after 3 to 4 hours, it shows a lower absolute error than Peleg's model, and this trend continues until the end.

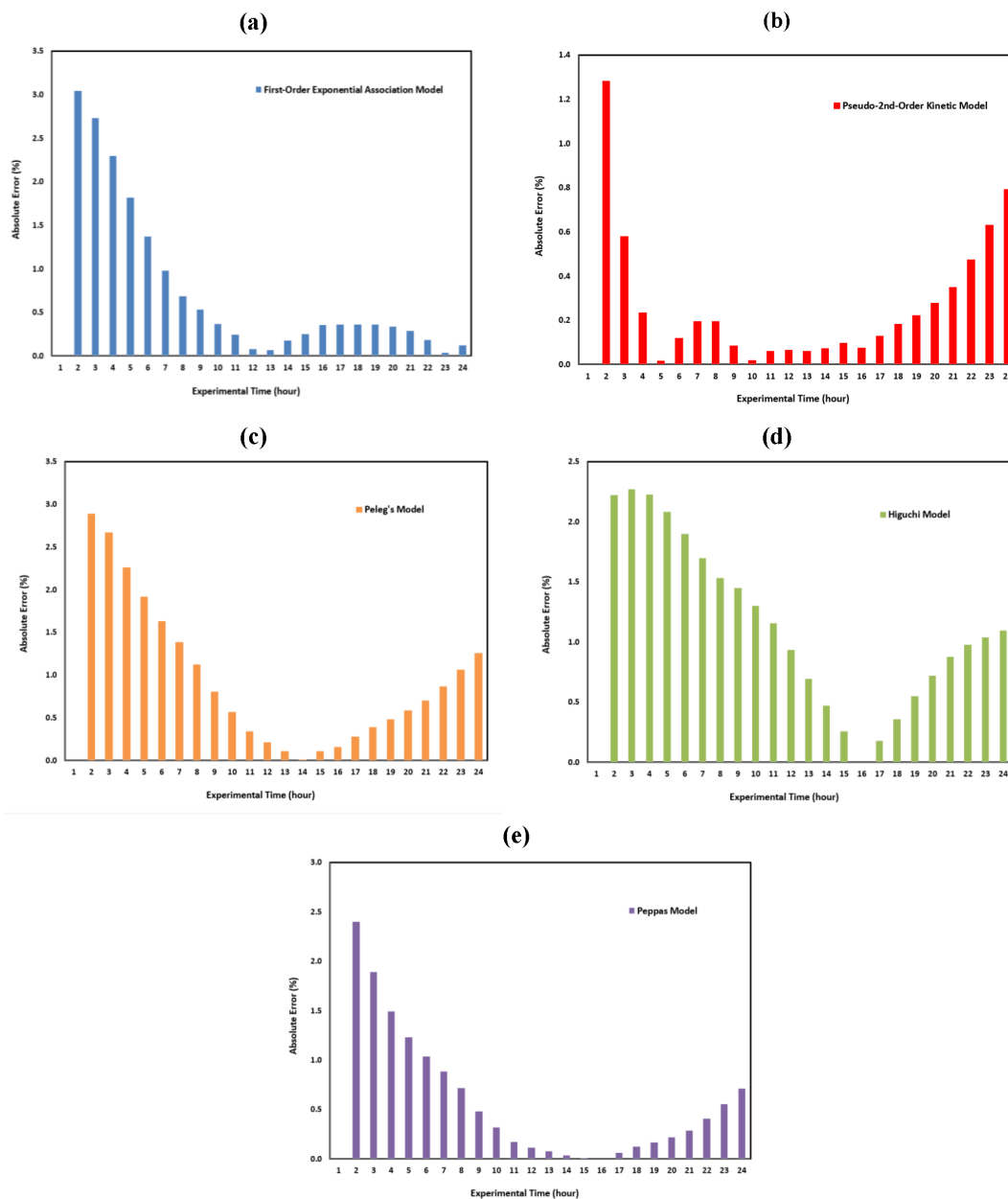


Figure 6—Absolute error varies with time. A first-order exponential association model, bPseudo-2nd-order kinetic model, cPeleg's model, dHiguchi model, and e Peppas model.

Figure 7 illustrates the mean absolute error (MAE) of different kinetic and diffusible models. It illustrates that the pseudo-2nd-order kinetic models have the lowest MAE value of 0.291%, indicating it is the best-performing model among all. In contrast, Higuchi and Peleg's models exhibit the highest MAE values, suggesting they deviate most from the experimental swelling data and are less effective in predicting the swelling behavior. The first-order exponential association model and Peppas model show MAE values of

0.738% and 0.609% respectively, indicating a moderate level of accuracy. The pseudo-2nd-order kinetic model is 52% better than the next best model (Peppas model) in terms of MAE. These findings demonstrate that the pseudo-2nd-order kinetic model is more reliable for modelling swelling behavior due to its superior performance compared to other models.

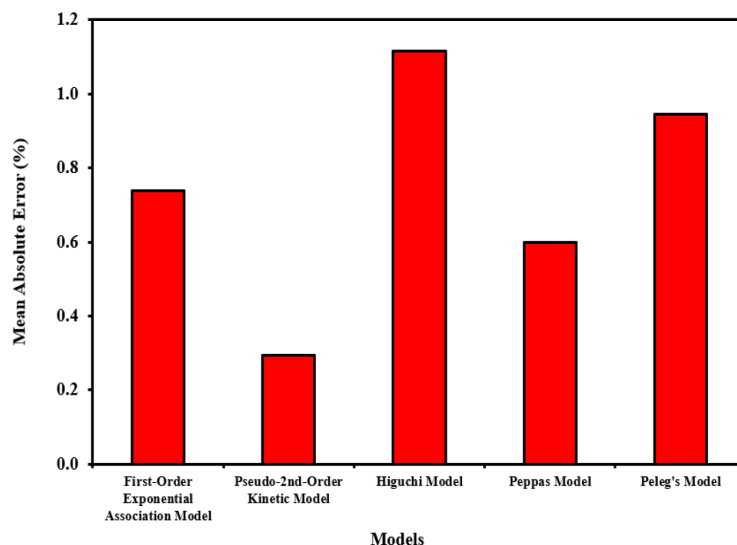


Figure 7—Mean Absolute Error (MAE) for all the models

Figure 8 Presents the absolute percentage relative error (APRE) values of each model, calculated using equation 13. APRE serves as a relative, normalized measure of prediction error concerning the experimental swelling values. Pseudo-2nd-order kinetic model demonstrates the lowest APRE values of 4.309%, corresponding to an accuracy of 96%, indicating the highest predictive accuracy among all models. In contrast, the Higuchi model records the highest APRE values of 15.55%, with the lowest accuracy of 84%, reflecting the poorest predictive model. The strong consistency between the MAE and APRE results further reinforces the conclusion that the pseudo-2nd-order kinetic model is the most reliable and accurate predictor of shale swelling behavior.

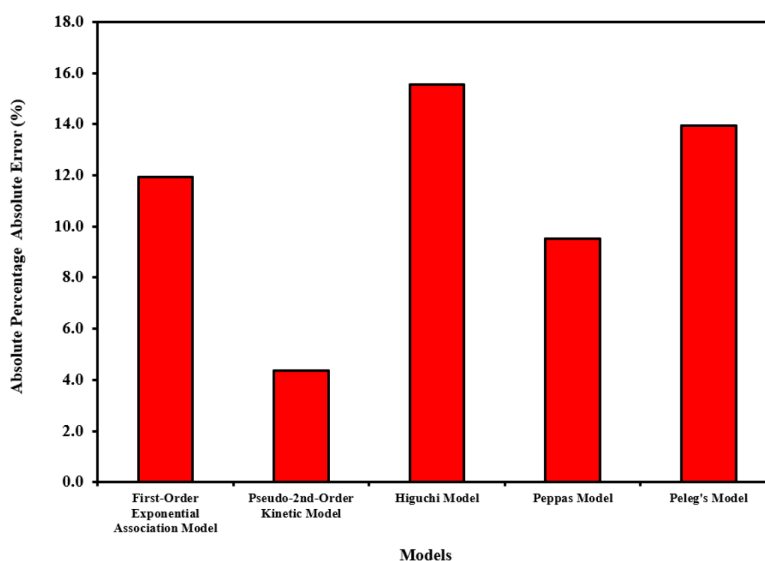


Figure 8—Absolute percentage relative error (APRE) for all the models.

Limitations

1. This study is based only on the performance of five selected models, not all.
2. The experimental swelling was evaluated using LDSM; no other field conditions or equipment were used.
3. The study focused only on the swelling behavior of WBDF, not on other types of fluids.
4. The results apply to the specific shale sample and may vary for other types of shale.

Conclusion

This study evaluates the comparison between predicted swelling models and the experimental swelling data obtained using the linear dynamic swelling meter (LDSM) test. The models used in this research study were the first-order exponential association model, pseudo-2nd-order kinetic model, Peleg's model, Higuchi model, and Peppas model. Among these, the pseudo-2nd-order kinetic model showed the best performance, highest predictive accuracy of 96% and the lowest MAE of 0.291% and APRE of 4.309%. The Higuchi Model exhibited the poorest performance, with the lowest predictive accuracy of 84% and the highest MAE of 1.116% and APRE of 15.55%. The first-order exponential association model, Peleg's model, and Peppas model demonstrate moderate predictive performance; however, the Peppas model performed slightly better, showing approximately 18% improvement over the first-order exponential association model in both MAE and APRE. Absolute error, relative error, and cross-plot consistently confirmed that the pseudo-2nd-order kinetic model is the most accurate and suitable model for characterizing shale swelling behavior in WBDF.

Future Recommendations

1. Machine learning-based regression (e.g., Random Forest, SVR) can be trained on the swelling dataset to predict complex behaviors, and application of nonlinear regression optimizers to improve prediction accuracy.
2. Conduct model fitting under variable pressures and temperatures, simulating actual wellbore conditions
3. Investigate how ionic strength, brine composition, and multivalent cations affect model performance
4. Incorporate depth-dependent swelling data to assess kinetic responses at specific depths
5. A software-based simulation tool could be built where users input shale composition, exposure time, temperature, etc., and obtain swelling predictions with each model.

Abbreviations

LDSM	Linear dynamic swelling meter
WBDF	Water-based drilling fluid
k_1, k_2, k	Characteristics rate constant
$S(t), S_t$	Swelling at a given time
M_t	Quantity of drug released over a specific time
S_e, q_e	Swelling at equilibrium
y_{exp}	Experimental swelling error
y_{pred}	Predicted swelling error
N	No of observation points
MAE	Mean absolute error
APRE %	Absolute percentage relative error

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